

DOCUMENT 05

User Flows & System Flows

Nine user flows, sequence diagrams, and seven state machines covering the full trade lifecycle.

SCOPE UF-01 to UF-09 • State Machines • Sequence Diagrams

SPONSOR Ghana Chamber of Gold Buyers

TECHNOLOGY PARTNER Vanta Technologies Ltd.

CONFIDENTIAL • CHAMBER & VENDOR INTERNAL • AUTHORIZATION DATE 15 AUGUST 2026

TABLE OF CONTENTS

Document 5 — User Flows & System Flows..... 3

5.1 User Flows..... 3

5.2 UF-03 — RFQ Submission & Negotiation (Detail).....3

 Decision Points & Edge Cases..... 4

5.3 UF-05 — Escrow Funding & Settlement (Detail).....4

 Edge Cases.....4

5.4 Sequence Diagram — RFQ Submission to Escrow Funding.....5

5.5 State Diagrams.....5

 Gold Lot State Machine..... 5

 Trade State Machine.....5

 Auction State Machine.....5

 Escrow State Machine..... 6

 Compliance Alert State Machine.....6

Right-click the table of contents and select "Update Field" to refresh page numbers after opening in Microsoft Word.

Document 5 — User Flows & System Flows

This document describes how users move through the product (screen-by-screen flows with branches and edge cases) and how their actions translate into system behavior (requests across services, sequence diagrams, state transitions). Each user flow is mapped to its primary persona, journey stage, and backing API endpoints. Nine user flows and seven state machines are covered.

5.1 User Flows

\#	FLOW	PRIMARY PERSONA	JOURNEY STAGE
UF-01	Member Onboarding	Ibrahim, Abena, Kwame	J1/2/3 Stage 1
UF-02	Lot Creation & Listing	Ibrahim	J3 Stage 2
UF-03	RFQ Submission & Negotiation	Kwame, Abena	J1/2 Stage 3
UF-04	Auction (Timed, Reverse, Sealed-Bid)	Kwame, Abena	J1/2 Stage 3 (alt)
UF-05	Escrow Funding & Settlement	Kwame, Abena, Ibrahim	J1/2/3 Stage 4
UF-06	Assay Request & Certification	Abena, Chamber	J2 Stage 3
UF-07	Export Documentation Bundle	Abena	J2 Stage 4
UF-08	Compliance Investigation	Efua	J4 Stage 2
UF-09	Dispute Resolution	All	Cross-journey

5.2 UF-03 — RFQ Submission & Negotiation (Detail)

The buyer browses Lots or receives a Tier 1 priority alert. On the lot detail page, the buyer submits an RFQ with an asking price (Tier 1 priority) or a bid (Tier 2 open). The seller is notified via WhatsApp + SMS. The seller accepts, counters, or rejects. Counter-offers iterate (max 5 rounds). On acceptance, a Trade is created (status PENDING_ESCROW). Both parties are notified with escrow funding instructions.

DECISION POINTS & EDGE CASES

DECISION POINT	LOGIC	UX IMPLICATION
Tier 1 priority alert (4h exclusivity)	Tier 1 buyers receive lot notifications 4h before Tier 2	Prominent countdown timer in negotiation UI
Counter-offer limits	Max 5 counter-offers per RFQ to prevent stalling	Counter after 5th rejected: maximum reached message
Auto-expiry	RFQ expires 24h after creation if not accepted	Seller sees countdown; buyer sees expires in X hours
High-value trade threshold	Trades $\geq$ US\$100K require MFA re-authentication	Please confirm MFA code to accept this trade above US\$100,000
Buyer not on seller's preferred panel	Warning shown but trade permitted	Subtle warning: This buyer is not on your preferred panel

5.3 UF-05 — Escrow Funding & Settlement (Detail)

Trade is created (status PENDING_ESCROW). Buyer taps Fund Escrow. If bank account not pre-authorized, mini onboarding. Confirm funding amount + bank account last4. If amount $\geq$ US\$100K, MFA challenge. POST /v1/trades/{id}/escrow/fund triggers bank adapter. Bank adapter calls partner bank API. On success, status FUNDED; both parties notified. Seller arranges armored transport via platform. Vault confirms receipt via operator API. Assay requested; certificate auto-attached. Title transfers on-platform. Escrow auto-released to seller bank account (or seller confirms receipt). Status SETTLED. Immutable audit entry + analytics event recorded.

EDGE CASES

EDGE CASE	HANDLING
Buyer's bank API is down	Bank adapter falls back to secondary partner bank (Charter Risk R2 mitigation); if both down, escrow marked BANK_PENDING, retry every 30 min for 4 hours
Vault receipt delayed $\geq$ 24h	Auto-alert to Chamber ops + seller + buyer; investigation workflow triggered
Assay result purity differs $\geq$ 0.5% from declared	Auto-pause settlement; both parties notified; renegotiation workflow triggered
Compliance hold placed during escrow	Escrow status FROZEN; trade status COMPLIANCE_HOLD; only Compliance Officer can release

EDGE CASE	HANDLING
Buyer disputes delivery (gold damaged/missing)	Trade status DISPUTED; escrow held; dispute workflow UF-09 triggered

5.4 Sequence Diagram — RFQ Submission to Escrow Funding

Buyer (Kwame) submits RFQ via PWA → API Gateway → RFQ Engine. RFQ Engine validates eligibility, writes RFQ aggregate + outbox row, publishes event "rfq.submitted" to Kafka. Notification Service consumes event and notifies Seller (Ibrahim) via WhatsApp + SMS. Seller submits counter-offer via PWA → Negotiation Service → DB write → Kafka event → WebSocket push to Buyer in real-time. Buyer accepts → RFQ Engine creates Trade via Trade Service → DB write (Trade aggregate, status PENDING_ESCROW) + outbox → Kafka event "trade.created" → Notification Service notifies both parties with escrow funding instructions. Buyer funds escrow → Escrow Service writes Escrow aggregate (PENDING) + outbox → calls Bank Adapter → Bank API. Bank responds 202 Accepted (async webhook will confirm funding). Bank later calls webhook → Escrow Service updates status to FUNDED → Kafka event "escrow.funded" → notifications + audit log.

5.5 State Diagrams

GOLD LOT STATE MACHINE

Lot lifecycle: DRAFT → LISTED → UNDER_OFFER → IN_NEGOTIATION → IN_ESCROW → DELIVERED → SETTLED. Side states: EXPIRED (7 days no offers), CANCELLED (seller withdraws), DISPUTED (dispute opened). A lot can transition from DISPUTED back to SETTLED (release) or to CANCELLED (refund).

TRADE STATE MACHINE

Trade lifecycle: PENDING_ESCROW → ESCROW_FUNDED → DELIVERED → ASSAYED → TITLE_TRANSFERRED → SETTLED. Side states: CANCELLED (buyer fails to fund within 24h), COMPLIANCE_HOLD (compliance freeze), DISPUTED. From COMPLIANCE_HOLD, can return to ESCROW_FUNDED (cleared) or transition to CANCELLED (refund). From DISPUTED, can transition to SETTLED (release) or CANCELLED (refund).

AUCTION STATE MACHINE

Auction lifecycle: SCHEDULED → LIVE → CLOSED → AWARDED → TRADE_CREATED. Side states: NO_SALE (reserve not met), CANCELLED (seller cancels with penalty), DISPUTED (winner fails to fund).

ESCROW STATE MACHINE

Escrow lifecycle: PENDING → FUNDED → RELEASED_TO_SELLER (or REFUNDED_TO_BUYER). Side states: FAILED (bank rejects funding), EXPIRED (24h no funding), FROZEN (compliance hold). From FROZEN, can transition to RELEASED_TO_SELLER (cleared) or REFUNDED_TO_BUYER (escalation to refund).

COMPLIANCE ALERT STATE MACHINE

Alert lifecycle: OPEN → INVESTIGATING → ESCALATED → FILED_SAR → MONITORED → CLEARED. Side transitions: INVESTIGATING → CLEARED (false positive); ESCALATED → CLEARED (council dismisses); MONITORED → ESCALATED (new pattern detected).